

Coordinate Transformations, “Fictitious Forces”, Christoffel Symbols, and All That

These notes should - ideally - provide some end-to-end understandable write-up of “accelerations observed in accelerating (i.e. non-inertial) coordinate systems.

The motivation for this write-up came from a desire to get a better grip on questions such as “how does one satellite in a constellation accelerate towards or away from another satellite” (where both “are not inertial in a Newtonian-Physics approach due to feeling the effect of gravity”), or, more subtly, “what is the Coriolis force contribution that’s attributable to our orbital plane precessing in a sun-synchronous orbit?”.

The derivations here are intentionally verbose, the idea being that at every point we should be able to relate intermediate quantities to the “velocities” and “accelerations” attributed to moving objects in different coordinate systems. At a purely algebraic level, the calculation could be shortened to less than one page by leaning more heavily into the chain rule, but that then would require more of an expert’s intuition to see how the expressions align with physical notions. So, rather than just doing Taylor expansion of the motion of an accelerating object subjected to a nonlinear coordinate transformation, we spend all the effort required to see the first nonlinear terms that affect observed accelerations for coordinate systems related by some nonlinear transformation. We will keep all summations explicit, and only take small and simple steps in algebraic manipulations.

This should - hopefully - keep explanations accessible to readers who have learned how to do some multivariate calculus, but do not have years of hands on experience with that kind of mathematics.

Some Basic Maths

We want to talk about coordinate transformations. In general, we will use (t, x, y, z) -coordinates: Even if we do not work with space-time in the relativistic sense, it simplifies the discussion a lot if we can describe “events” as “this time and that position” coordinate-4-tuples.

Let us consider having two different “coordinate systems”, i.e. two different ways to attribute a 4-tuple of numbers to some given event E . Let’s use capital letters for “System A”-coordinates and small letters for “System B” coordinates. Loosely:

$$E@A = (T, X, Y, Z), \quad E@B = (t, x, y, z)$$

In general, we will have functions that express $TXYZ$ -coordinates in terms of $txyz$ -coordinates and vice-versa, and whenever we need to look at local coordinate-transformations at a point, we can think of the relevant functions to be described using either the $TXYZ$ -coordinates or $txyz$ -coordinates. In many

situations, one choice is more convenient, and it pays to think a little about what approach works better. We will come to this.

Let's consider a simple situation: We are given a trajectory of an object (such as: a satellite) in terms of a 4-coordinate-tuple valued function of some curve-parameter τ , which we here can take to be time. So, for position-coordinates in the A -coordinate-system, we have: $P @ A(\tau) = (T(\tau), X(\tau), Y(\tau), Z(\tau))$, i.e. $P^0(\tau) @ A = T(\tau)$, $P^1(\tau) = X(\tau)$ etc.

Two notes:

- We start index counting at zero, and index-zero is our time-coordinate. This is aligned with common conventions in relativity, and makes our explorations carry over directly from Newtonian to Einsteinian gravitational physics.

This is nothing to be scared about - rather, we should think of this as “let's align with the highly streamlined and systematic notation and approach used by people who have to analyze way hairier problems than everything we are going to encounter” rather than “let's stick with the idea to not use any maths beyond late-19th-century-engineering mathematics as much as possible”. *We wouldn't want to do this with Roman Numerals either.*

- One perhaps slightly strange-looking consequence here is that we have a curve-parameter (above: τ) that we can take to be simply “wall clock time”. This gives us, for the time-coordinate of any coordinate system we want to use: $T(\tau) = t(\tau) = \tau$. Could we not then simplify all this by dropping this 4th coordinate “time” and staying 3d? At the conceptual level, the situation closely parallels the one with 3d computer graphics that is generally done with 4d projective coordinates: using the 4d approach simplifies the structure of the mathematics a lot, since we get “time/space coordinate mixing contributions” that otherwise would make conceptually very simple expressions have additional summands.

Multi-dimensional Taylor Expansion

Let's say we have some analytic function of one variable, such as perhaps $f(x) = \exp(-x^2/2)$. “Being analytic” means that if we expand the function into a Taylor series at a given point x_0 , then at every point x that is less than the radius of convergence away from x_0 , that series will converge to the original function's value. It is by no means true that all functions are analytic, i.e. that there is some neighborhood of x_0 inside which the Taylor series always converges to the value of the original function, but for the problems we are looking into, we can nevertheless take this to always hold.

We know we have - using somewhat informal writing (“you see how the full sum will look like”):

$$f(x_0 + h) = f(x_0) + (h^1/1!)f'(x_0) + (h^2/2!)f''(x_0) + (h^3/3!)f'''(x_0) + \dots$$

The good news: “Physics” (mostly) makes us care only about positions, velocities, and accelerations, not higher derivatives. (Sometimes, the 3rd derivative, the “jerk” - or rate-of-change of force, respectively, acceleration - will play a role. Sudden onset/stopping of a force, such as when releasing a ball from a sling, would be “infinite jerk”.)

So, we will here be able to truncate such expansions at the second derivative.

The bad news: We right away have to ponder how the above expansion generalizes from $\mathbb{R}^1 \rightarrow \mathbb{R}^1$ functions to $\mathbb{R}^n \rightarrow \mathbb{R}^n$ functions - here specifically with $n = 4$.

More concretely, let’s say we have a way to express the T, X, Y, Z coordinates in terms of t, x, y, z -coordinates. The information-content of such a $\mathbb{R}^4 \rightarrow \mathbb{R}^4$ function is equivalent to having a vector of four $\mathbb{R}^4 \rightarrow \mathbb{R}^1$ functions, one per coordinate.

So, if for example, (t, x, y, z) are coordinates for an “inertial” (not-accelerating, “free fall”) coordinate system where objects not experiencing any force will move with constant velocity, and the (T, X, Y, Z) coordinate system shares the same origin, but rotates around the z -axis with angular velocity ω , we could describe this as:

$$\begin{aligned}\tilde{T}(t, x, y, z) &= t \\ \tilde{X}(t, x, y, z) &= x \cos(\omega t) - y \sin(\omega t) \\ \tilde{Y}(t, x, y, z) &= x \sin(\omega t) + y \cos(\omega t) \\ \tilde{Z}(t, x, y, z) &= z\end{aligned}$$

Already here, we are seeing why it makes sense to treat t as a coordinate on the same footing. We are using the “tilde” above the name of each coordinate to indicate that these are not coordinates (numbers) but functions - but this is not some widely established convention.

Let’s use the name X^i for the entries of the vector-of-functions $(\tilde{T}, \tilde{X}, \tilde{Y}, \tilde{Z})$. A few things are somewhat unfortunate here:

- While it is customary in this context to place vector-coordinate indices “upstairs”, this can cause some ambiguity w.r.t. exponent-or-index. In our explorations here, we will never encounter an exponent on a coordinate, resolving this question.
- We have a name clash X^i (the collection of four coordinate-functions) and the coordinate $X = X^1$.

- There is a natural isomorphism between $p \mapsto (\tilde{T}(p), \tilde{X}(p), \tilde{Y}(p), \tilde{Z}(p))$, which “has type $\mathbb{R}^4 \rightarrow \mathbb{R}^4$ ” and X^i as introduced above, where the function-tuple “has type $(\mathbb{R}^4 \rightarrow \mathbb{R}, \mathbb{R}^4 \rightarrow \mathbb{R}, \mathbb{R}^4 \rightarrow \mathbb{R}, \mathbb{R}^4 \rightarrow \mathbb{R})$ ”, and going forward, we will just liberally switch between these different pictures without saying so.

The latter is a source of potential confusion that makes learning these things unnecessarily hard, and it totally is a consequence of a culturally widely established lack of effort to be precise in constructions.

So, how does the multi-dimensional Taylor expansion of the coordinate-transformation function look like? As we said, we only need to go to 2nd order.

$$X^i(p+h) = X^i(p) + (1/1!) \sum_m \frac{\partial X^i}{\partial x^m} h^m + (1/2!) \sum_{m,n} \frac{\partial^2 X^i}{\partial x^m \partial x^n} h^m h^n + \{\text{higher order terms}\}$$

Here, $\frac{\partial X^i}{\partial x^m}$ is a matrix of partial derivatives, “taking the derivative of every capital-letter coordinate-function by every small-letter coordinate” which then gets evaluated at the point with small-letter coordinates p (this is silent/implicit). Likewise for the quadratic term. So, if $Y(t, x, y, z) = X^2(t, x, y, z) = y - 7x^3z^5 - 2tx$, then $\frac{\partial Y}{\partial x \partial z} = \frac{\partial X^2}{\partial x^1 \partial x^3} = 105x^2z^4$.

Above, it is noteworthy that the quadratic term that has a 2nd partial derivative which is a 3-index object: the totality of these derivatives is indexed by (i, m, n) . This is not a “matrix”, but (equivalent to) a 3-index-array of coefficients - perhaps best thought of not as some sort of 3d coordinate scheme, but a relational database table with one “value” column and three index-columns. Here, we already see how multiindex objects arise very naturally and “simple matrix/vector linear algebra doesn’t quite cut it”.

Body Motion

Our next step is about considering a moving body. Let’s make $\tau = t - t_0$ “time measured from some given reference point in time”. We only care about position, velocity, and acceleration, so the momentary situation around $\tau = 0$ can be described as follows:

$$x^i(\tau) = \hat{x}^i + \hat{v}^i \tau + \frac{1}{2} \hat{a}^i \tau^2 + \{\text{higher order terms we do not care about}\}$$

Here, $(\hat{x}^i, \hat{v}^i, \hat{a}^i)$ are the position-coordinates, velocity-coordinates, and acceleration-coordinates for the momentary position, velocity, and acceleration at $\tau = 0$, expressed in the (small-letters) B -coordinate-system. Clearly, this again is simply a Taylor expansion which we truncated to 2nd order.

The next step is pretty clear now: We need to substitute this “object motion to 2nd order in time” into the coordinate-transformation that gets us X^i -coordinates in terms of x^m -coordinates, which we also expand to 2nd order in coordinates. Since we are only interested in accelerations in the X^i -coordinates, we can straight away drop all terms $\propto \tau^3, \propto \tau^4, \dots$

Let us consider $x^i(\tau) = \hat{x}^i + h_\tau^i$, so $h_\tau^i = \hat{v}^i \tau + \frac{1}{2} \hat{a}^i \tau^2$ and substitute into what we had above:

$$X^i(p+h) = X^i(p) + \sum_m \frac{\partial X^i}{\partial x^m} h_\tau^m + \frac{1}{2} \sum_{m,n} \frac{\partial^2 X^i}{\partial x^m \partial x^n} h_\tau^m h_\tau^n + \{\dots\}$$

Expanding:

$$\begin{aligned} X^i(p_\tau) = X^i(p+h_\tau) &= \\ &X^i(p) \\ &+ \sum_m \frac{\partial X^i}{\partial x^m} (\hat{v}^m \tau + \frac{1}{2} \hat{a}^m \tau^2) \\ &+ \frac{1}{2} \sum_{m,n} \frac{\partial^2 X^i}{\partial x^m \partial x^n} (\hat{v}^m \tau + \frac{1}{2} \hat{a}^m \tau^2) (\hat{v}^n \tau + \frac{1}{2} \hat{a}^n \tau^2) + \{\dots\} \end{aligned}$$

Now, we note that from the quadratic term, we can only get a $\propto \tau^2$ -term if a $\hat{v}^m$ -coordinate multiplies a $\hat{v}^n$ -coordinate. Everything else is $\propto \tau^3, \propto \tau^4$. Collecting contributions by power-of- τ , we get:

$$X^i(p_\tau) = X^i(p+h_\tau) = X^i(p) + \tau \cdot \sum_m \frac{\partial X^i}{\partial x^m} \hat{v}^m + \frac{1}{2} \tau^2 \cdot \left(\sum_m \frac{\partial X^i}{\partial x^m} \hat{a}^m + \sum_{m,n} \frac{\partial^2 X^i}{\partial x^m \partial x^n} \hat{v}^m \hat{v}^n \right) + \{\dots\} \quad (*)$$

This already looks interesting:

1. The position in new coordinates we get by simply using the coordinate-transformation functions on the position in old coordinates.
2. The velocity in new coordinates is simply the velocity in old coordinates, transformed with the coordinate-transformation matrix.
3. The acceleration-vector in new coordinates is simply the velocity-vector in old coordinates, coordinate-transformed to the new coordinate system with the “how much change in the new 2nd coordinate per change in the old 3rd coordinate” type weighting “that relates X^i -change to x^m -change”, *plus a new term that somehow involves “the bending of the X -coordinates expressed in the x -coordinates”.*

We will look into especially (3.) in more detail in a moment, but it is worthwhile to ponder (2.) a bit: Let’s consider the very simple situation where x^i -coordinates describe coordinates “relative to spaceship in interstellar space, no thrust, inertial

motion”, and X^i describe “coordinates relative to a probe released by the spaceship. Same orientation of the coordinates, but $X^i = x^i - b^i t = x^i - b x^0$ ”. Commonly, we would want to think about velocities as satisfying $V^i = v^i - b^i$. Note that there are *two summands* here, while in our above treatment, we only have a *single* matrix-vector product! Indeed, if we were to work out how this matrix looks like, we would get *exactly* the same situation that we know from 3-d computer graphics in projective coordinates (as alluded to earlier): The $\partial T/\partial t = 1$ -entry makes it possible to absorb the shift into matrix multiplication!

The relation to Christoffel Symbols

Above, we found how to express the X^i -coordinate-acceleration in terms of coordinate-transformed x^i -coordinate-acceleration (as we would expect), plus “some other term that depends on velocities and gives us accelerations if the relation between old and new coordinates is not linear”.

Overall, this expression is pretty useful. For example, if we know the acceleration of one satellite relative to another satellite in Cartesian coordinates, but want to know the angle-acceleration (since this will translate to some motor force) for a telescope whose pointing direction we would like to describe in satellite-fixed (latitude, longitude) coordinates, we would simply use this formula to convert to (time, latitude, longitude, distance)-coordinates and could immediately read off angle-accelerations.

Physics: Centrifugal and Coriolis Forces In many applications, both coordinate systems will be such that the spatial directions are Cartesian, but “these Cartesian coordinate systems can move and perhaps rotate in time”. Then, the “velocity-transformation” gets us just the $\vec{v} \mapsto M\vec{v} + \vec{w}$ rotation-plus-shift transformation we see in a “purely 3d” treatment.

How about the “acceleration” terms? Here, it is noteworthy that, since we always have $v^0 = 1$ in our approach, we get $a^0 = 0$, and hence there is no such “shift”-term for the time-coordinate akin to the shift-terms for spatial coordinates provided by the vector $\vec{w}$ above for the velocity-transformation. For the 2nd term, the one with the “second partial derivative”, if the relation between spatial coordinates is linear-with-perhaps-an-offset (i.e. “affine”), *as we have it if spatial coordinates at any time are Cartesian* (where we then only are free to rotate and shift, for the given point in time), if both derivatives are w.r.t. spatial coordinates, the corresponding $\partial^2/\partial\partial$ -coefficient is zero.

We hence only get “interesting forces” if one or both the partial derivatives involve the time-coordinate.

Let’s briefly specialize to a rotating coordinate system where time only shows up in the coordinate-transformation formulae via $\sin(\omega t)$ and $\cos(\omega t)$ -factors. In that case, a nontrivial first time-derivative will come with one power of ω , and a second time-derivative will come with a factor ω^2 . Note that for centrifugal

acceleration, we have $a = r\omega^2$. Indeed, this “double time derivative” contribution from this term gives us the centrifugal force.

What about the “mixed” term with one spatial and one time-derivative? This will give us an acceleration-contribution that is proportional to ω , and also to the spatial part of $\hat{v}^m$ (the other v^2 -factor will here be $v^0 = 1$). We also get a factor-2 from “either v -factor can be the v^0 ”. This is the Coriolis force: In a rotating system, an object moving at velocity $\vec{v}$ will experience a “fictitious force” acceleration $2\vec{v} \times \vec{\omega}$, where $\vec{\omega}$ is the rotation-vector (pointing along the axis of rotation, length being the angular velocity.)

It makes sense to try to link this to intuition. Considering a tall tower at the equator, if we drop a rock from the top of the tower, it accelerates towards the ground. Will it land to the east or to the west of the point right underneath where it was dropped (whose position we could determine with a plumb line)? Pondering the situation in a non-rotating coordinate system, the rock at the top of the tower is further away from earth’s rotation axis than the bottom of the tower, so it is moving faster eastward than the bottom. As it falls, it will retain that excess eastward velocity, which will make it land further to the east. The “velocity excess” is directly proportional to the vertical distance fallen, and seen as the eastward velocity by a ground-fixed rotating observer. Distance fallen is proportional to the square of the time passed since the rock was dropped, so the *rate-of-change* of eastward velocity is proportional to the time that has passed since dropping the rock - but so is downward velocity. So, the most relevant contribution to eastward acceleration (very small) comes from downward velocity, and - unsurprisingly - also is proportional to earth’s rotation rate, $|v_{east}| \propto |\omega| \cdot |v_{down}|$. Also, the right hand rule shows that the cross product of a “downward” vector with the rotation-axis vector (towards the North Star) indeed points eastward, in alignment with $\vec{a}_{\text{Coriolis}} = 2\vec{v} \times \vec{\omega}$.

Finally: The Connection Coefficients

Let us now consider the most general situation. As we just have seen, changing to a different coordinate system can give rise to velocity-dependent accelerations.

In the generic case, *the “original” B -coordinate system already comes with such velocity-dependent accelerations*, and it is natural to expect that we simply have to coordinate-transform these to get a contribution to the (coordinate-)acceleration in the “new” coordinate system - but there will be another contribution that is related to the “new” coordinates “accelerating relative to the ‘old’ coordinates”.

We hence want to consider not one, but a collection of different trajectories in the B -coordinate system, and we describe these in terms of accelerations that we understand as being a sum of two contributions,

$$\hat{a}^m = \alpha^m - \sum_{n,p} \gamma^m_{np} \hat{v}^n \hat{v}^p \quad (**) \quad$$

where $\hat{v}^n$ is the velocity in B -coordinates. So, the idea here is that an observer who describes the world in the B -coordinate-system, which in general can be a “non-inertial” coordinate system, might regard an acceleration they observe as the sum of some acceleration that an inertial observer would see (in that observer’s C -coordinate-system), but coordinate-transformed to their own coordinate system - that would be the α^m term above - plus some velocity-dependent acceleration-contributions that encode (for example) the Coriolis force. So, from the perspective of an observer using B -coordinates, if two bodies have *the same* acceleration in B -coordinates $\hat{a}^m$, but one is at rest while the other one moves with some B -coordinates velocity, then they would have different accelerations in C -coordinates.

Now, what would the acceleration in A -coordinates then be? We can simply read off that acceleration A^i from the term multiplying $\frac{1}{2}\tau^2$ in Eq. (*):

$$A^i = \sum_m \frac{\partial X^i}{\partial x^m} \hat{a}^m + \sum_{m,n} \frac{\partial^2 X^i}{\partial x^m \partial x^n} \hat{v}^m \hat{v}^n$$

Now, let us substitute in the $\hat{a}^m$ from above:

$$A^i = \sum_m \frac{\partial X^i}{\partial x^m} \left(\alpha^m - \sum_{n,p} \gamma^m_{np} \hat{v}^n \hat{v}^p \right) + \sum_{m,n} \frac{\partial^2 X^i}{\partial x^m \partial x^n} \hat{v}^m \hat{v}^n$$

Rearranging and relabeling summation indices:

$$A^i = \sum_m \frac{\partial X^i}{\partial x^m} \alpha^m + \sum_{n,p} \left(\sum_m - \frac{\partial X^i}{\partial x^m} \gamma^m_{np} \hat{v}^n \hat{v}^p + \frac{\partial^2 X^i}{\partial x^n \partial x^p} \hat{v}^n \hat{v}^p \right) \quad (***)$$

What is noteworthy here is that Eq. (**) expressed the velocity-dependent acceleration-corrections in B -coordinates that get added to the C -coordinates-accelerations-transformed-to- B -coordinates α^m in terms of the B -coordinates velocities, while in the expression above, we do almost the same thing, but the velocities we are referring to are B -coordinate velocities. We would like these to be A -coordinate velocities instead. We know how these velocities are related - the A -velocities are the term multiplying τ in Eq. (*):

$$V^i = \sum_m \frac{\partial X^i}{\partial x^m} \hat{v}^m.$$

Since $\left(\frac{\partial X}{\partial x}\right)^{-1} = \left(\frac{\partial x}{\partial X}\right)$, or, more precisely,

$$\sum_m \left(\frac{\partial X^i}{\partial x^m} \right) \left(\frac{\partial x^m}{\partial X^j} \right) = \delta_j^i,$$

we have:

$$\hat{v}^m = \sum_i \frac{\partial x^m}{\partial X^i} V^i$$

So, substituting this velocity-expression into (***) gets us:

$$A^i = \sum_m \frac{\partial X^i}{\partial x^m} \alpha^m + \sum_{n,p} \left(- \sum_m \frac{\partial X^i}{\partial x^m} \gamma^m_{np} + \frac{\partial^2 X^i}{\partial x^n \partial x^p} \right) \left(\sum_j \frac{\partial x^n}{\partial X^j} V^j \right) \left(\sum_k \frac{\partial x^p}{\partial X^k} V^k \right)$$

Let us rearrange and simplify a bit:

$$A^i = \sum_m \frac{\partial X^i}{\partial x^m} \alpha^m + \sum_{m,n,p} \sum_{j,k} - \left(\gamma^m_{np} \frac{\partial X^i}{\partial x^m} \frac{\partial x^n}{\partial X^j} \frac{\partial x^p}{\partial X^k} \right) V^j V^k + \sum_{n,p} \sum_{j,k} + \frac{\partial^2 X^i}{\partial x^n \partial x^p} \frac{\partial x^n}{\partial X^j} \frac{\partial x^p}{\partial X^k} V^j V^k$$

Remember that in B -coordinates, we had, for velocity-dependent acceleration terms (Eq. (**)),

$$\hat{a}^m = \alpha^m - \sum_{n,p} \gamma^m_{np} \hat{v}^n \hat{v}^p$$

and so if we likewise want this to hold in A -coordinates in the form:

$$\hat{A}^i = \sum_m a^m \frac{\partial X^i}{\partial x^m} - \sum_{j,k} \Gamma^i_{jk} V^j V^k,$$

then comparing these expressions tells us that we want this to hold:

$$-\Gamma^i_{jk} = \sum_{m,n,p} \left(-\gamma^m_{np} \frac{\partial X^i}{\partial x^m} \frac{\partial x^n}{\partial X^j} \frac{\partial x^p}{\partial X^k} \right) + \sum_{n,p} \frac{\partial^2 X^i}{\partial x^n \partial x^p} \frac{\partial x^n}{\partial X^j} \frac{\partial x^p}{\partial X^k}$$

Let us rearrange this by multiplying with the appropriate inverse matrices to extract the γ^m_{np} . (This might look a bit scary, but if we think about it, it is a straightforward step: For every “open index” that occurs at the left hand side, we coordinate-transform with the “value of a small linear coordinate update in one system expressed in the currency of the other system’s small linear coordinate updates” matrix. Three indices require three such matrices. The formula looks complicated since we are looking at a “wiring diagram” that should be seen as a graph but was rendered as a mathematical term via typography instead. This is a bit like trying to describe a reaction in organic chemistry by not showing molecules, but their names only.)

$$\begin{aligned} \sum_{m,n,p,i,j,k} \gamma^m_{np} \left(\frac{\partial X^i}{\partial x^m} \frac{\partial x^n}{\partial X^j} \frac{\partial x^p}{\partial X^k} \right) \left(\frac{\partial x^{m'}}{\partial X^i} \frac{\partial X^j}{\partial x^{n'}} \frac{\partial X^k}{\partial x^{p'}} \right) = \\ = \sum_{i,j,k} \Gamma^i_{jk} \left(\frac{\partial x^{m'}}{\partial X^i} \frac{\partial X^j}{\partial x^{n'}} \frac{\partial X^k}{\partial x^{p'}} \right) + \sum_{i,j,k} \sum_{n,p} \frac{\partial^2 X^i}{\partial x^n \partial x^p} \frac{\partial x^n}{\partial X^j} \frac{\partial x^p}{\partial X^k} \left(\frac{\partial x^{m'}}{\partial X^i} \frac{\partial X^j}{\partial x^{n'}} \frac{\partial X^k}{\partial x^{p'}} \right) \end{aligned}$$

This leaves us with:

$$\gamma^{m'}_{n'p'} = \sum_{i,j,k} \Gamma^i_{jk} \left(\frac{\partial x^{m'}}{\partial X^i} \frac{\partial X^j}{\partial x^{n'}} \frac{\partial X^k}{\partial x^{p'}} \right) + \sum_i \frac{\partial^2 X^i}{\partial x^{n'} \partial x^{p'}} \frac{\partial x^{m'}}{\partial X^i}$$

Or, after renaming indices:

$$\gamma^m_{np} = \sum_{i,j,k} \Gamma^i_{jk} \left(\frac{\partial x^m}{\partial X^i} \frac{\partial X^j}{\partial x^n} \frac{\partial X^k}{\partial x^p} \right) + \sum_i \frac{\partial^2 X^i}{\partial x^n \partial x^p} \frac{\partial x^m}{\partial X^i}.$$

This is the known expression for expressing the “velocity-dependent accelerations-corrections” of one coordinate system (here the B -coordinate system) in terms of that of another coordinate system.

This then is the “transformation law for Christoffel Symbols (of the 2nd kind)”. If we did not include the 2nd summand - the one with the 2nd partial derivatives - this would correspond to coordinate-transforming the “parallel-transport connection for in-general-not-Cartesian coordinate system A” so that we get the “parallel-transport connection for coordinate system A, expressed in the coordinates of coordinate system B”. The 2nd term turns that quantity into the “parallel-transport connection for coordinate system B, expressed in the coordinates of coordinate system B”, or shorter, “connection for coordinate system B”. Notably, if the “parallel-transport connection for coordinate system A” is nonzero, it will always be nonzero irrespective of what coordinate system we use to describe it (since the coordinate transformations are invertible). The 2nd, “inhomogeneous” term can make them zero for one coordinate system while they are nonzero for another. Indeed, it is always possible to locally make the Γ^i_{jk} zero by switching to the coordinate system of some (non-rotating) free-falling observer. This gives us “Riemann Normal Coordinates”.